
Project Documentation

Project Name

Linux Web Server using Nginx on Ubuntu

Objective

Build and manage a Linux web server using Nginx and access it from another machine on the network.

Environment

Host Machine

Windows

Virtual Machine

Ubuntu Linux

Hypervisor

VMware Workstation

Web Server

Nginx

Phase 1: Installation

Updated package repository:

```
sudo apt update
```

Installed Nginx:

```
sudo apt install nginx
```

Verified service:

```
sudo systemctl status nginx
```

Phase 2: Service Management

Started Nginx:

```
sudo systemctl start nginx
```

Enabled automatic startup:

```
sudo systemctl enable nginx
```

Learned difference:

enable = start at boot

start = start now

Phase 3: Network Verification

Checked IP address:

```
ip addr
```

Found:

192.168.154.132

Verified connectivity:

```
ping
```

Phase 4: Port Verification

Checked listening services:

```
sudo ss -tulnp | grep :80
```

Observed:

0.0.0.0:80

Meaning:

Nginx listening on all IPv4 interfaces

Phase 5: Custom Web Page

Modified website files:

```
/var/www/html
```

Created custom page:

Welcome to Venky's Linux Web Server

Verified browser access.

Phase 6: Firewall Configuration

Installed and configured UFW:

```
sudo ufw allow 80
```

Checked rules:

```
sudo ufw status
```

Phase 7: Log Analysis

Monitored access logs:

```
sudo tail -f /var/log/nginx/access.log
```

Learned:

Every browser request generates a log entry.

Phase 8: Troubleshooting Incident

Problem

Windows Host could not access:

```
http://192.168.154.132
```

Error:

Unable to connect to remote server

Investigation Process

Step 1

Verified network:

```
ping 192.168.154.132
```

Result:

Success

Conclusion:

Network connectivity exists.

Step 2

Verified firewall:

```
sudo ufw status
```

Result:

Port 80 allowed

Step 3

Verified Nginx listening:

```
sudo ss -tulnp | grep :80
```

Result:

Port 80 listening

Step 4

Captured packets:

```
sudo tcpdump -nn -i ens33 'host 192.168.154.1 and tcp port 80'
```

Observed:

Windows → SYN
Ubuntu → RST

Meaning:

Connection actively rejected.

Step 5

Checked service status:

```
sudo systemctl status nginx
```

Found:

Active: inactive (dead)

Root Cause

Nginx service was stopped.
Since no process was listening on port 80:

Ubuntu returned TCP RST packets.
This caused:

Windows could not connect.

Resolution

Started service:

```
sudo systemctl start nginx
```

Verified:

```
sudo systemctl status nginx
```

Result:

Active: active (running)

Final Verification

From Windows:

```
curl http://192.168.154.132
```

Result:

HTTP 200 OK

Custom webpage displayed successfully.

Skills Learned

Linux

- Service management
- File system navigation
- Log analysis

Networking

- IP addressing
- Ports
- TCP handshake
- Packet capture

Web Server

- Nginx installation
- Nginx management
- Static website hosting

Troubleshooting

- Root cause analysis
 - Firewall validation
 - Packet-level debugging
 - Service verification
-

Final Grade

If this were an interview lab:

Deployment : PASS

Networking : PASS

Troubleshooting : PASS

Documentation : PASS

Project 1: Linux Web Server with Nginx - Completed ✓

Now don't immediately run off to Kubernetes because your brain saw the word "server" and wants to learn the next 47 technologies. Spend one day making sure you can repeat this project from scratch without notes. That's how knowledge turns into skill. 😊